

Topic 01: The Big Picture, Order, Linearity & Verifying Solutions

Professor Leonard Master Series · Lessons 1 to 4

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Professor Leonard's Whiteboard Intuition: What is a Differential Equation?

Imagine you are driving a car down the highway.

- In **Algebra**, someone asks:

"Where are you right now?"

You give a static number:

"Mile marker 50."

- In **Calculus 1**, someone asks:

"How fast are you changing your position right now?"

You give a derivative:

"65 miles per hour."

- In **Differential Equations**, someone asks:

"If your speed at any moment is proportional to where you are, where will you be in two hours?"

A Differential Equation is simply a mathematical sentence that relates an **unknown function** $y(x)$ to its **rates of change** (y' , y'' , $\dots$). When you "solve" an algebraic equation like $x^2 - 4 = 0$, the answer is a **number** ($x = \pm 2$). When you "solve" a differential equation like $y' = 2x$, the answer is a **whole family of functions** ($y(x) = x^2 + C$).

2. Order, Degree & Linearity (Leonard's Rules of the Road)

Rule 1: The Order is King

Look for the highest derivative on the board:

$$y''' + 4(y')^5 - y = 0 \implies \text{Order } 3$$

The exponent 5 on y' is the **degree**, but the **order** is strictly 3.

Rule 2: Linearity is Fragile

Professor Leonard explains linearity like this:

"The dependent variable y and its derivatives are delicate. The moment you do anything fancy to them, linearity breaks!"

- If you square $y \implies$ **Nonlinear** (y^2)
- If you take the cosine of $y \implies$ **Nonlinear** ($\cos y$)
- If you multiply y by its own derivative $\implies$ **Nonlinear** ($y \cdot y'$)
- If y appears in an exponent $\implies$ **Nonlinear** (e^y)

Only terms like $x^3 y'' + \sin(x) y' + e^x y = \ln(x)$ are **linear**, because the coefficients involve only the independent variable x , while y, y', y'' stand alone to the first power.

3. How to Verify a Solution (The 3-Step Whiteboard Protocol)

Example: Verify that $y(x) = C_1 e^{2x} + C_2 e^{-3x}$ is a solution to $y'' + y' - 6y = 0$.

Step 1: Compute the necessary derivatives

$$y = C_1 e^{2x} + C_2 e^{-3x}$$

$$y' = 2C_1 e^{2x} - 3C_2 e^{-3x}$$

$$y'' = 4C_1 e^{2x} + 9C_2 e^{-3x}$$

Step 2: Plug directly into the Left-Hand Side (LHS)

$$\begin{aligned} \text{LHS} &= y'' + y' - 6y \\ &= (4C_1 e^{2x} + 9C_2 e^{-3x}) + (2C_1 e^{2x} - 3C_2 e^{-3x}) - 6(C_1 e^{2x} + C_2 e^{-3x}) \end{aligned}$$

Step 3: Collect like terms

$$e^{2x} \text{ terms : } (4C_1 + 2C_1 - 6C_1)e^{2x} = 0e^{2x} = 0$$

$$e^{-3x} \text{ terms : } (9C_2 - 3C_2 - 6C_2)e^{-3x} = 0e^{-3x} = 0$$

$$\text{LHS} = 0 = \text{RHS} \quad \checkmark$$

The relation holds for all values of C_1 and C_2 !

4. Leonard's Red Flag Alerts

-  **Red Flag:** Treating an Initial Value Problem condition $y(0) = 4$ as a point $x = 4, y = 0$. Remember: the inside is x , the outside is y ! $y(x_0) = y_0 \implies (x_0, y_0) = (0, 4)$.